WHAT SECTION 6 GIVES AT SLOPE ± 2

Outcome. I could not derive $S^3_2(K) \ncong S^3_{-2}(K)$ from Section 6, even with nondegeneracy hypotheses. Section 6 does give an exact formula for every drop, with the window ends as corollaries. The formula shows precisely what is missing: the position of the class realising the spectral invariant. I see no hypothesis of transversality type that controls this position, and the data show that it is not controlled for free.

1. Set-up

$K \subset S^3$ satisfies (H1)–(H3) and $\Delta''_K(1) = 0$. The latter is forced by a cosmetic pair through the Casson–Walker invariant. Let $U = (0, \frac{1}{2})_a \times (\mathbb{R}/\mathbb{Z})_b$ be the pillowcase minus the edges $a \in \{0, \frac{1}{2}\}$, with $\omega = 2\,da \wedge db$. By Proposition 1.1(b), $\Gamma = \Gamma_K \subset U$. The surgery lines $L^\varepsilon_N = \{Na + b \equiv \varepsilon/2\}$ are arcs crossing U. Since Γ misses the edges, all these arcs are interchangeable as far as Γ is concerned.

Proposition 1 (from Corollary 6.4; minimal position assumed)**.** *$CI(Y_{\pm 2}, \varepsilon) \cong CI(Y_{\pm 1}) \cong CI(Y_0, w)$ as $\mathbb{Z}/8$-graded complexes, up to grading shifts, for both ε. The S^3 terms vanish. In particular there is no unfiltered obstruction: the problem lives entirely in the Chern–Simons data. Moreover Γ has total winding 0 around U. By Herald's formula (§3.1), the algebraic count of $\Gamma \cap L_{1/n}$ is $2\lambda(Y_{1/n}) = n\Delta''_K(1) = 0$.*

2. The exact drop formula

Let $\lambda_{1/4} = 2(a - \frac{1}{4})\,db$. This primitive of ω is invariant under $\chi(a, b) = (\frac{1}{2} - a, -b)$. Let Φ be its primitive along Γ, which is $d\Phi = 2(a - \frac{1}{4})db$ of `status.pdf`. On L_N the restriction of $\lambda_{1/4}$ is $d\big(-N(a - \frac{1}{4})^2\big)$. With the line primitive normalised to vanish at its traceless point $a = \frac{1}{4}$, the pillowcase level formula reads

$$c(Y_N, x) = A_N(x) + \tfrac{3}{16}\operatorname{sgn} N - \tfrac{N}{16} - \tfrac{1}{8}\iota_N(x), \qquad A_N(x) = \Phi(x) + N\big(a(x) - \tfrac{1}{4}\big)^2.$$

This is exact for torus knots and checked numerically for 4_1. For $N = \pm 2$ the constant is $\pm\frac{1}{16}$: this is where the sixteenth comes from.

Proposition 2 (rotation identity)**.** *Let $P = (a_P, b_P)$ be a common point of L^ε_2 and $L^{\varepsilon'}_{-2}$. Let L_t, $t \in [-2, 2]$, be the lines through P of slope $-t$, and let $x_t \in \Gamma \cap L_t$ move continuously, with no birth or death. Then the level moves at the rate $-(a_t - a_P)^2 - a_P(\frac{1}{2} - a_P)$. Hence*

$$c(Y_2, x_2) - c(Y_{-2}, x_{-2}) = g_P - \int_{-2}^{2} \big(a(x_t) - a_P\big)^2 dt - \tfrac{1}{8}\Delta\iota,$$

where $g_P = \frac{1}{8}$ for the same bundle ($P = (\frac{1}{4}, \frac{\varepsilon - 1}{2})$) and $g_P = \frac{3}{16}$ for mixed bundles ($a_P \in \{\frac{1}{8}, \frac{3}{8}\}$).

Proof. On $\{t(a - a_P) + b - b_P = 0\}$ we have $\dot b = -(a - a_P) - t\dot a$. Hence $\frac{d}{dt}A_t = 2(a - \frac{1}{4})\dot b + (a - \frac{1}{4})^2 + 2t(a - \frac{1}{4})\dot a$. The terms in $\dot a$ cancel, leaving $(a - \frac{1}{4})(2a_P - a - \frac{1}{4})$. Adding the $-\frac{1}{16}$ of the constant gives the stated rate. The constant jumps by $\frac{3}{8}$ at $t = 0$. At $x = P$ the integral vanishes and the gap is g_P. □

This is Section 6's $\delta(x)$ written as a swept area. For a corner pivot it is literally Theorem 6.3: the triad $(1, \frac{1}{0}, 2)$ has midpoint $m_1 = (\frac{1}{2}, \frac{1}{2})$, a corner. *Check:* for $T(2, 3)$ the identity the level formula gives $u = \frac{3}{32}$ and $w = \frac{1}{4}$, as in the tables. Here Γ is the segment $b = \frac{1}{2} - 6a$ and $\Phi = -6(a - \frac{1}{4})^2$. This is a weak check: the formula was calibrated on torus knots.

Corollary 3 (windows)**.** *Assume continuity of spectral invariants along the family (standard for the holonomy perturbations of §5.3), and that the grading term $\Delta\iota$ vanishes along it, as in every example. Then every drop equals g_P minus the integrated squared distance of the* spectral carrier *from P. Hence the same-bundle drop lies in $(-\frac{1}{8}, \frac{1}{8}]$ and the mixed drop in $(-\frac{3}{8}, \frac{3}{16}]$. The upper ends are attained exactly when the carrier sits at the common point.*

These are the known windows: DLME's map H and the map B for the same bundle, and the third map for mixed bundles. The equality cases are the common-point cases already seen in the data.

3. Why this does not prove the theorem

A cosmetic pair needs every drop to vanish: bottom and top, both bundles. By the identity this means that each carrier's integrated squared distance from its pivot equals g_P. For the same bundle, $\int_{-2}^{2}(a_t - \frac{1}{4})^2 dt = \frac{1}{8}$: an RMS distance of $\frac{1}{4\sqrt{2}}$ from traceless. Nothing in Section 6 excludes this:

- Γ is an arbitrary exact, χ-invariant multicurve of winding 0 in U;
- the cone isomorphisms only transport generators along such rotations;
- transversality, minimal position and distinct levels say nothing about where the carrier sits.

The data show that it is not controlled for free. For positive torus knots the trivial-bundle drop u is negative, so that carrier's deficit exceeds $\frac{1}{8}$.

A hypothesis bounding the carrier's distance from P does suffice. For example, if $\Gamma \subset \{|a - \frac{1}{4}| < \frac{\sqrt{2}}{8}\}$, every same-bundle drop is positive. But such a bound is a bound on the Hamiltonian along the carrier, i.e. an energy hypothesis in disguise. This is the input the no-go theorem said was missing: *where the surviving class sits.*

4. Where a lower bound can come from

Lower bounds do not come from rotations through the horizontal, i.e. through $Y_{\pm 1}$, Y_0 and the cones of Section 6. They can only come from the other route, through the traceless circle $\{a = \frac{1}{4}\}$: the bundle-exchanging triangle with the theory of (S^3, K), which is not a Farey triad. There a class that is not absorbed has mixed drop greater than $\frac{1}{8}$ (the kernel criterion of `triangle-1`, conditional on its hypotheses). Absorption is the only failure mode, and Section 6 says nothing about it. Zero-energy absorption needs a common traceless point with $\lambda \mapsto \pm 1$. For $\Delta_K = 1$ these are not binary dihedral and occur only degenerately, as for $P(-3, 5, 7)$.

5. Calibration at ± 1, where the answer is known

L_1 and L_{-1} meet only at the corners $(0, 0)$ and $(\frac{1}{2}, \frac{1}{2})$. Section 6 joins $Y_{\pm 1}$ through (Y_0, w) by the triads $(0^w, \frac{1}{0}, \pm 1)$, which is rotation about the corner $C = (\frac{1}{2}, \frac{1}{2})$. The same computation gives

$$\ell(Y_1) - \ell(Y_{-1}) = \tfrac{3}{8} - \mathrm{Area}(T) - \tfrac{1}{8}\Delta\iota,$$

where T is the region swept by the carrier, with vertex C. *Check:* for the right-handed trefoil the level formula gives $\frac{1}{80} - (-\frac{23}{112}) = \frac{61}{280}$, which is DLME's $\frac{59}{120} - \frac{23}{84}$; here $\Delta\iota = 0$ and $\mathrm{Area}(T) = \frac{43}{280}$. The window $(-\frac{3}{8}, \frac{3}{8})$ contains 0, so *Section 6 does not prove the* ± 1 *case either.*

DLME's lower bound $\ell(Y_1) - \ell(Y_{-1}) \geq \frac{1}{8} + \eta$ comes from their map $g_1 \colon Y_1 \to Y_{-1}$: a one-parameter family with $b^+ = 0$ and level $-\frac{1}{8} - E$, which is an isomorphism because $I(S^3) = 0$. Take a pair $x \mapsto y$ that g_1 counts, with x, y on one component of Γ. Stokes gives

$$E = \tfrac{1}{4} - \mathrm{Area}(T_{xy}) - \tfrac{1}{8}\Delta\iota.$$

So positivity of g_1's energy *is* the area bound that the cones cannot see. At ± 2 the map in g_1's role is B, of level $+\frac{1}{8}$. Positivity of its energy gives only a drop greater than $-\frac{1}{8}$.